Capstone Expo

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected capstone evidence record.

Assemble the bundle, reconcile counts, repair claims and name the next evidence need.

Keep one habit: Retain data, contract, code version, full traces, tests, failures, support and unresolved limits.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Improve and Challenge. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Build the Recommender. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Improve and Challenge

Build the Recommender

Agents and Tools

W1 · Assemble reproducible evidence

> List inputs, contract, code, versions, results and failure history.

Your evidence record

- Activity contract for this level: nonempty unique names; minutes exactint0..60;fun/learning exactint0..5; Boolean not integer; validate before selecting; eligibleminutes<=limit; score2fun+learning; highest score then ascending name; invalid rows retained with reasons; distinguish invalid request and no eligible option. Pictured book program uses different fields and score2interest+illustrated with ID ties.

- List the files or records a peer needs to reproduce the chooser result.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Check report consistency

> Reconcile supplied, accepted and rejected counts.

Your evidence record

- Activity contract for this level: nonempty unique names; minutes exactint0..60;fun/learning exactint0..5; Boolean not integer; validate before selecting; eligibleminutes<=limit; score2fun+learning; highest score then ascending name; invalid rows retained with reasons; distinguish invalid request and no eligible option. Pictured book program uses different fields and score2interest+illustrated with ID ties.

- A report says supplied 5, accepted 4 and rejected 2. Identify the inconsistency.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair an educational-benefit claim

> State the bounded software claim actually demonstrated.

Your evidence record

- Activity contract for this level: nonempty unique names; minutes exactint0..60;fun/learning exactint0..5; Boolean not integer; validate before selecting; eligibleminutes<=limit; score2fun+learning; highest score then ascending name; invalid rows retained with reasons; distinguish invalid request and no eligible option. Pictured book program uses different fields and score2interest+illustrated with ID ties.

- A presentation says the project proves students learn better. What claim does a successful rule-based demo support instead?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Name the next evidence need

> Do not infer universal correctness or learning benefit from one pass.

Your evidence record

- Activity contract for this level: nonempty unique names; minutes exactint0..60;fun/learning exactint0..5; Boolean not integer; validate before selecting; eligibleminutes<=limit; score2fun+learning; highest score then ascending name; invalid rows retained with reasons; distinguish invalid request and no eligible option. Pictured book program uses different fields and score2interest+illustrated with ID ties.

- After passing the new case, what remains unproven and what is a reasonable next step?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Defend a new activity case independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Activity contract for this level: nonempty unique names; minutes exactint0..60;fun/learning exactint0..5; Boolean not integer; validate before selecting; eligibleminutes<=limit; score2fun+learning; highest score then ascending name; invalid rows retained with reasons; distinguish invalid request and no eligible option. Pictured book program uses different fields and score2interest+illustrated with ID ties.

Limit is 10. A has minutes 10, fun 2, learning 4; B has 9,3,2; C has 11,5,5; D has 8,"4",2. Validate, rank, give counts, explain the winner, then reverse the **input** order. Present without hints.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh **transfer**.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Defend a new case independently

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Final **evidence bundle**: brief, data, scoring logic, tests, improvement record, and product card.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Check report consistency for Defending the data application. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Defending the data application. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Present a recommender that gets 8/10 final requests right. One failure invents a title; another violates a hard page limit. Is '**80%**, ready for everyone' justified? Propose release criteria.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

An expo claim says the recommender knows everybody's favourite book after twelve tests.

Rewrite the defence.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Your demo recommends the right book once. An audience member asks why they should trust it. Name **evidence** and one **limitation** you should present.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 36 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Present a recommender that gets 8/10 final requests right. One failure invents a title; another violates a hard page limit. Is '**80%**, ready for everyone' justified? Propose release criteria.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Limit is 10. A has minutes 10, fun 2, learning 4; B has 9,3,2; C has 11,5,5; D has 8,"4",2. Validate, rank, give counts, explain the winner, then reverse the **input** order. Present without hints.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

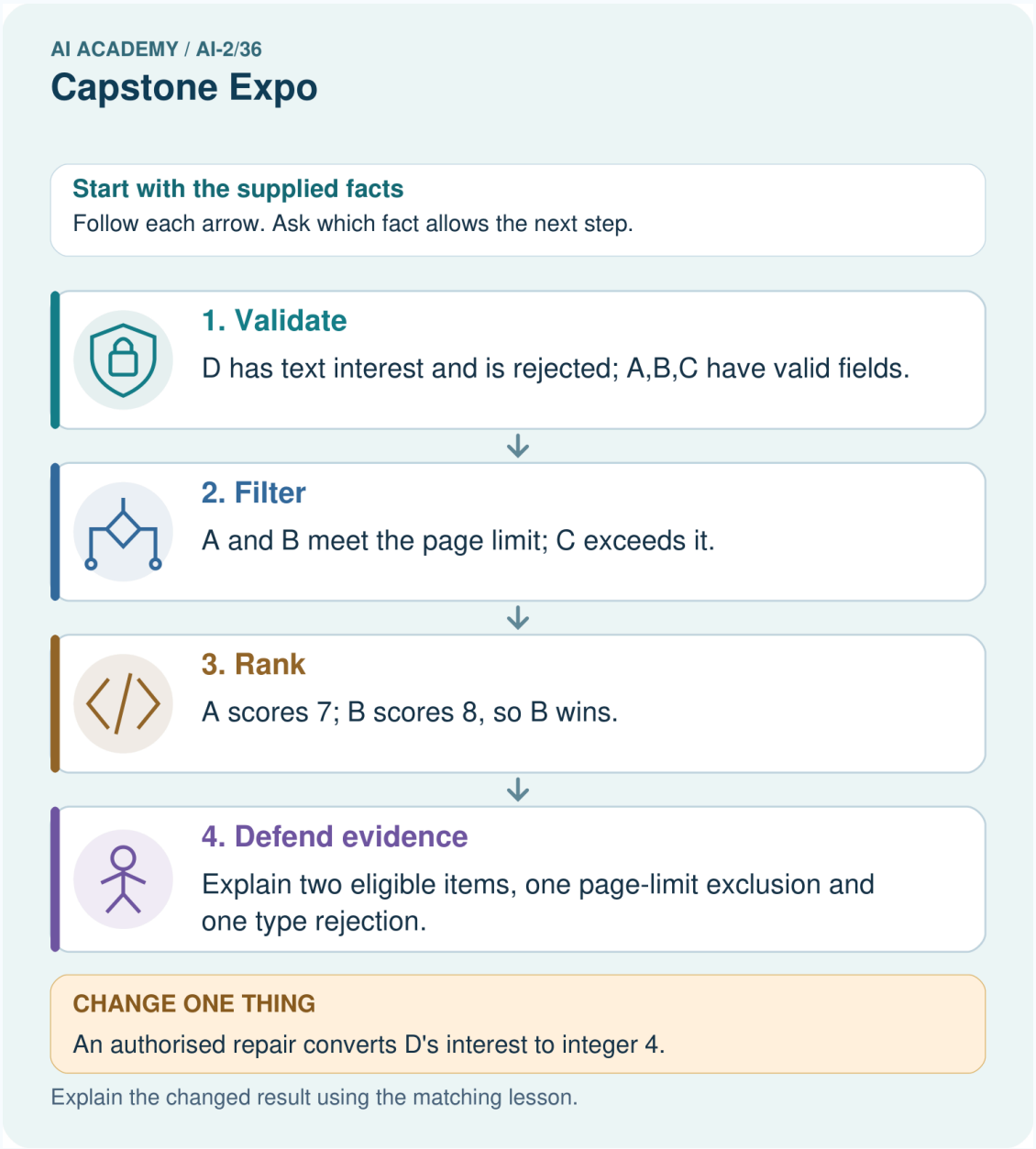
> Keep the explanation and the support you used.

Use this when helpful

A fictional team presents its transparent book recommender on a supplied practice catalogue.

Request maximum 120 pages; score=2×interest+illustrated, where illustrated is 1 for True and 0 for False; tie by ID.

A pages120 interest3 True; B pages100 interest4 False; C pages121 interest5 True; D pages90 interest"4" True. Numeric fields require exact integers.



Follow the labelled picture with the matching explanation. Open the vector diagram

Give the original winning ID, its score, and counts of eligible, excluded and invalid rows.

> **My picture-based explanation and changed-case prediction**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Present the Transparent Book Recommender's code, baseline, full-case traces, reserved results and **limitation**s, without claiming fictional results as pupil work.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Present the Transparent Book Recommender's code, baseline, full-case traces, reserved results and **limitation**s, without claiming fictional results as pupil work.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional team presents its transparent book recommender on a supplied practice catalogue.
- Which labelled step matches this detail? Explain two eligible items, one page-limit exclusion and one type rejection.
- 1. Defend **evidence**

- 2. Validate
- 3. Filter
- 4. Rank
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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> | |
> |---|
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> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Defend **evidence** Explain two eligible items, one page-limit exclusion and one type rejection. A presentation should let someone rebuild the result from the stated contract. A's equality is accepted; D is not silently converted; C's high preference score cannot override the page requirement. Reversing rows must preserve B because it has the unique highest eligible score. This pictured case is supported practice; keep the separately named independent activity task for a first attempt before feedback.

Changed-case question

- Changed-case practice: An authorised repair converts D's interest to integer 4. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.

- 3. D becomes eligible and scores 9, so it wins.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: D becomes eligible and scores 9, so it wins. The **evidence**-backed **input** repair changes validation and then ranking; record this as a new case/version.

Repair and save your work

Review Improve and Challenge (ai-2/35). Goal: Repair a specific observed failure. Short practice: Why are tests at both 100 and 101 needed?. Return to this week and explain what you changed.

Review Build the Recommender (ai-2/33). Goal: Implement a deterministic recommender. Short practice: Explain why B does not win after D is excluded even though it has fewer pages..

Return to this week and explain what you changed.

Review Agents and Tools (ai-2/30). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Can the assistant accept Trace B because its final state is done?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
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> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

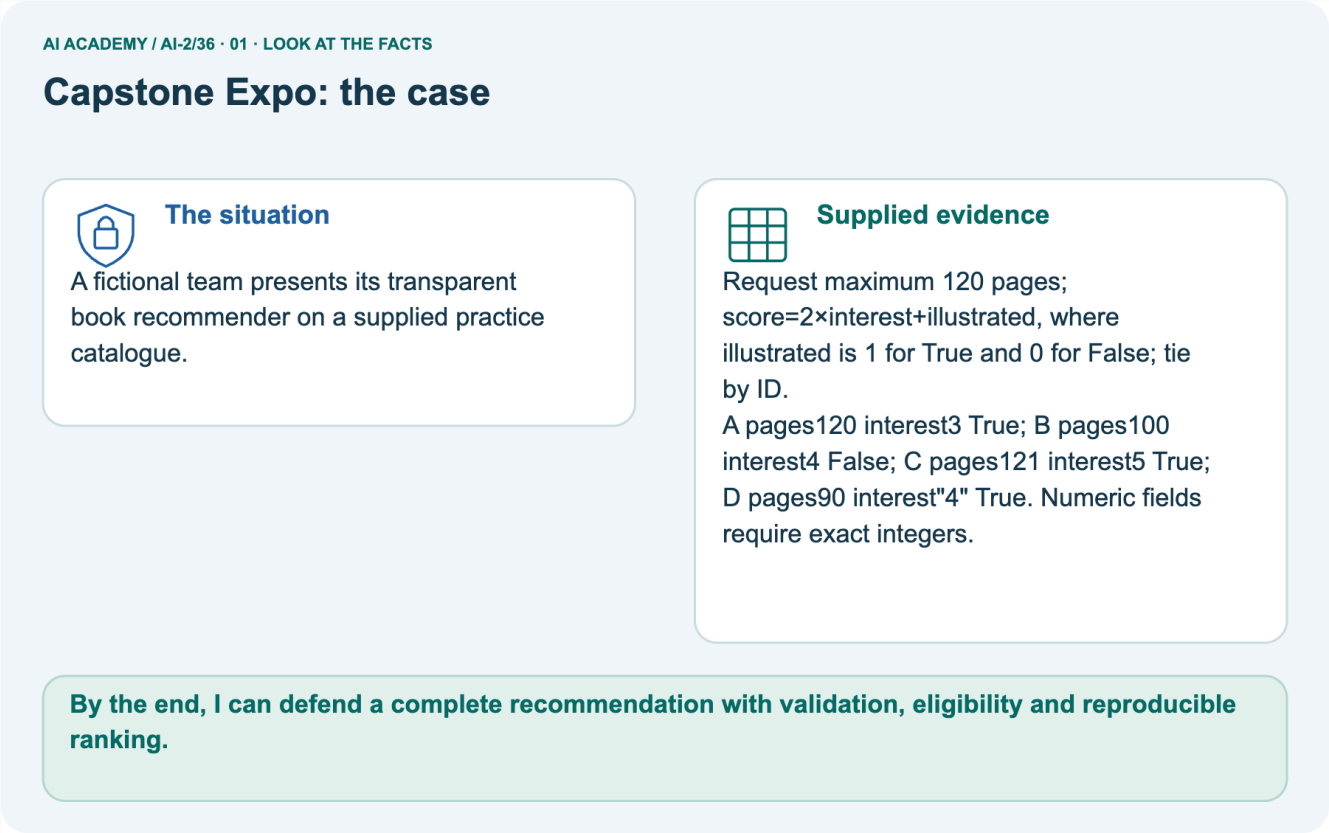
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Capstone Expo: the case



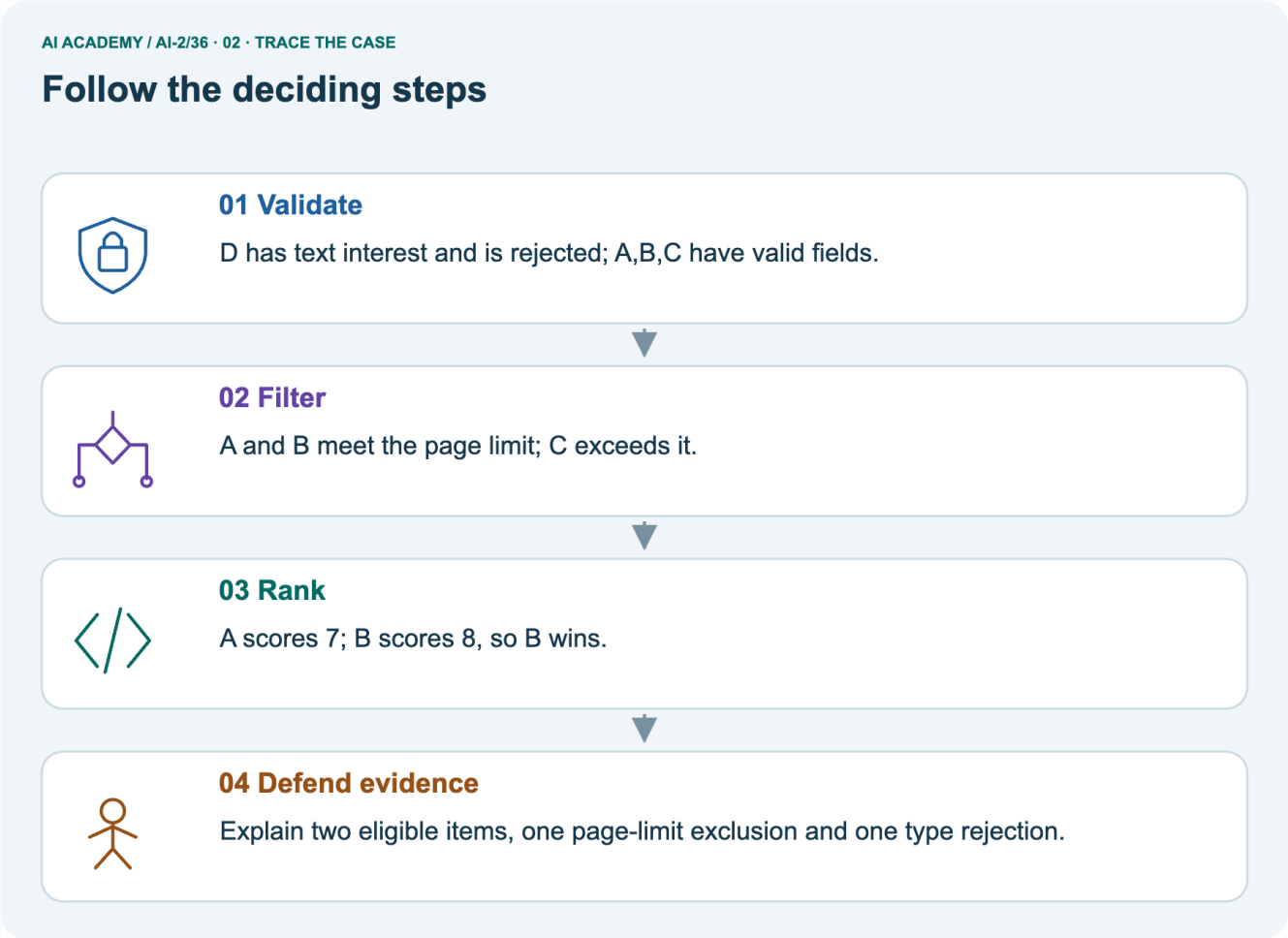
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional team presents its transparent book recommender on a supplied practice catalogue. Supplied facts: Request maximum 120 pages; score=2×interest+illustrated, where illustrated is 1 for True and 0 for False; tie by ID. A pages120 interest3 True; B pages100 interest4 False; C pages121 interest5 True; D pages90 interest"4" True. Numeric fields require exact integers.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



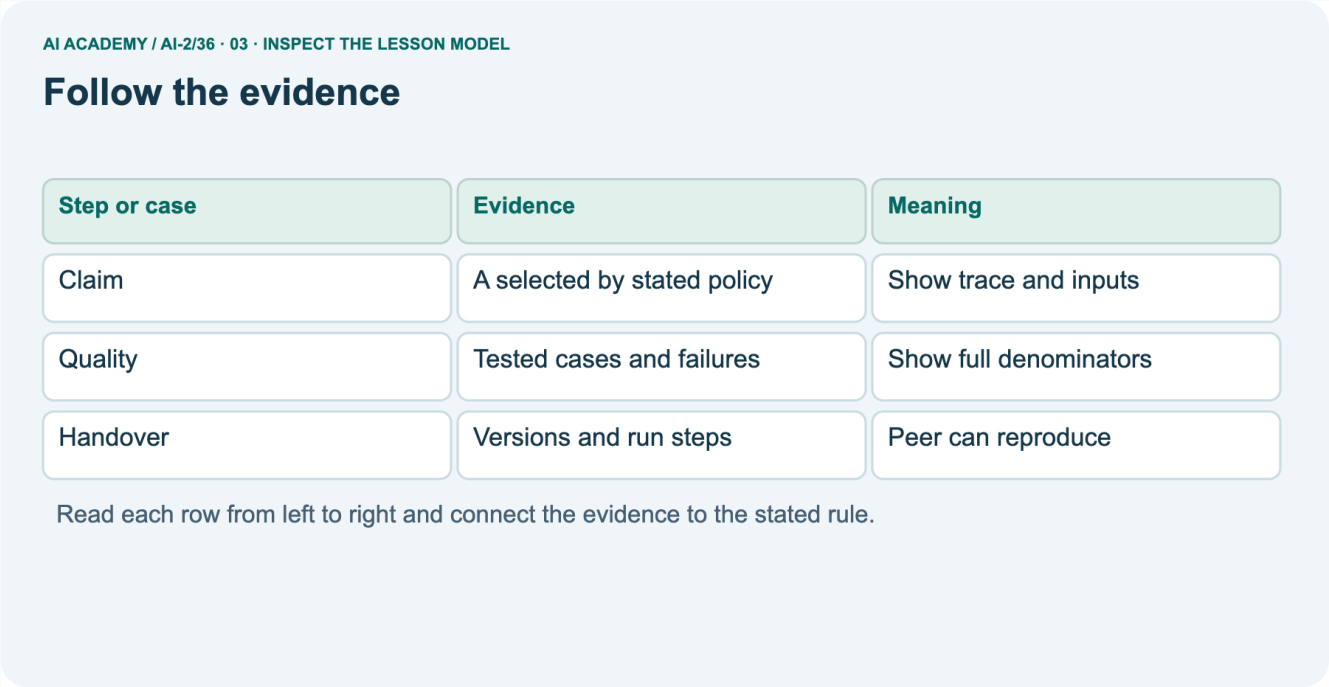
Follow the labelled picture with the matching explanation. Open the vector diagram

Validate: D has text interest and is rejected; A,B,C have valid fields. → Filter: A and B meet the page limit; C exceeds it. → Rank: A scores 7; B scores 8, so B wins. → Defend **evidence**: Explain two eligible items, one page-limit exclusion and one type rejection.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence



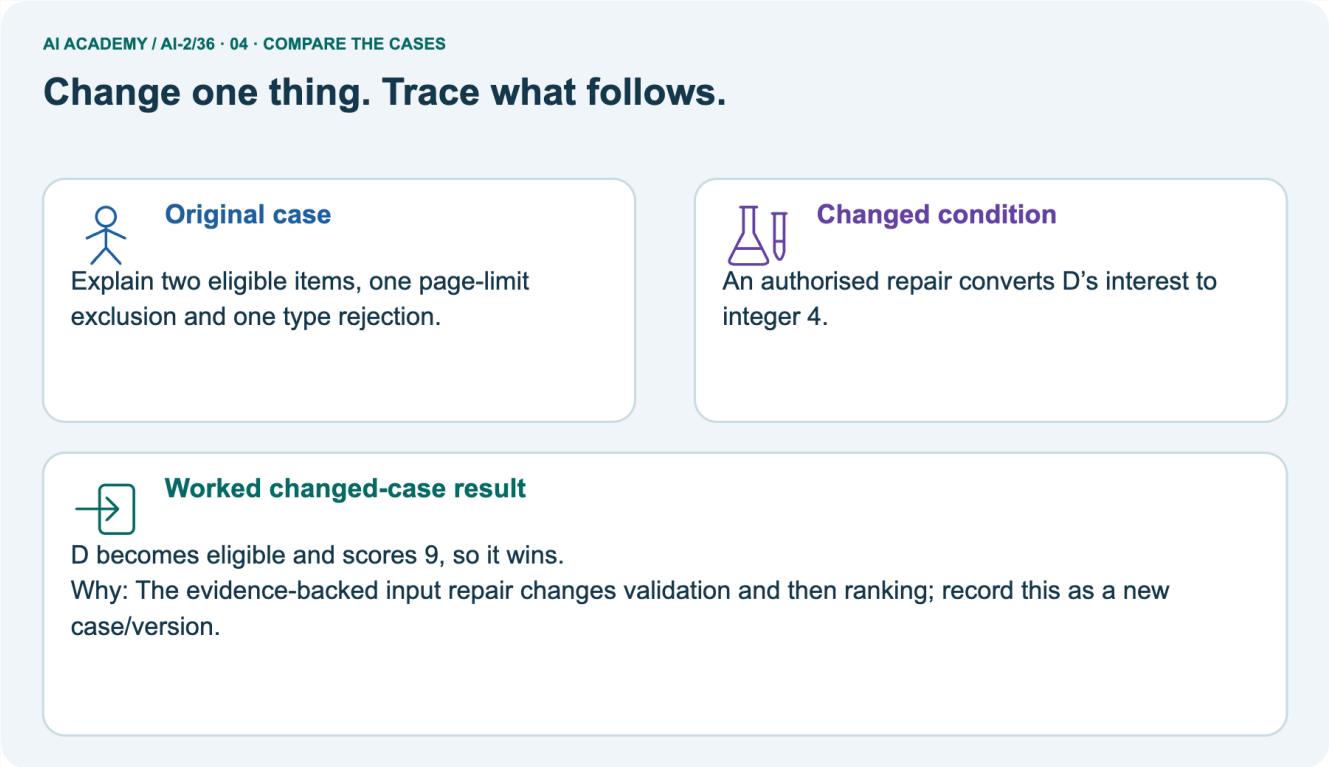
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Claim | A selected by stated policy | Show trace and **input**s. Quality | Tested cases and failures | Show full denominators. **Handover** | Versions and run steps | Peer can reproduce

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

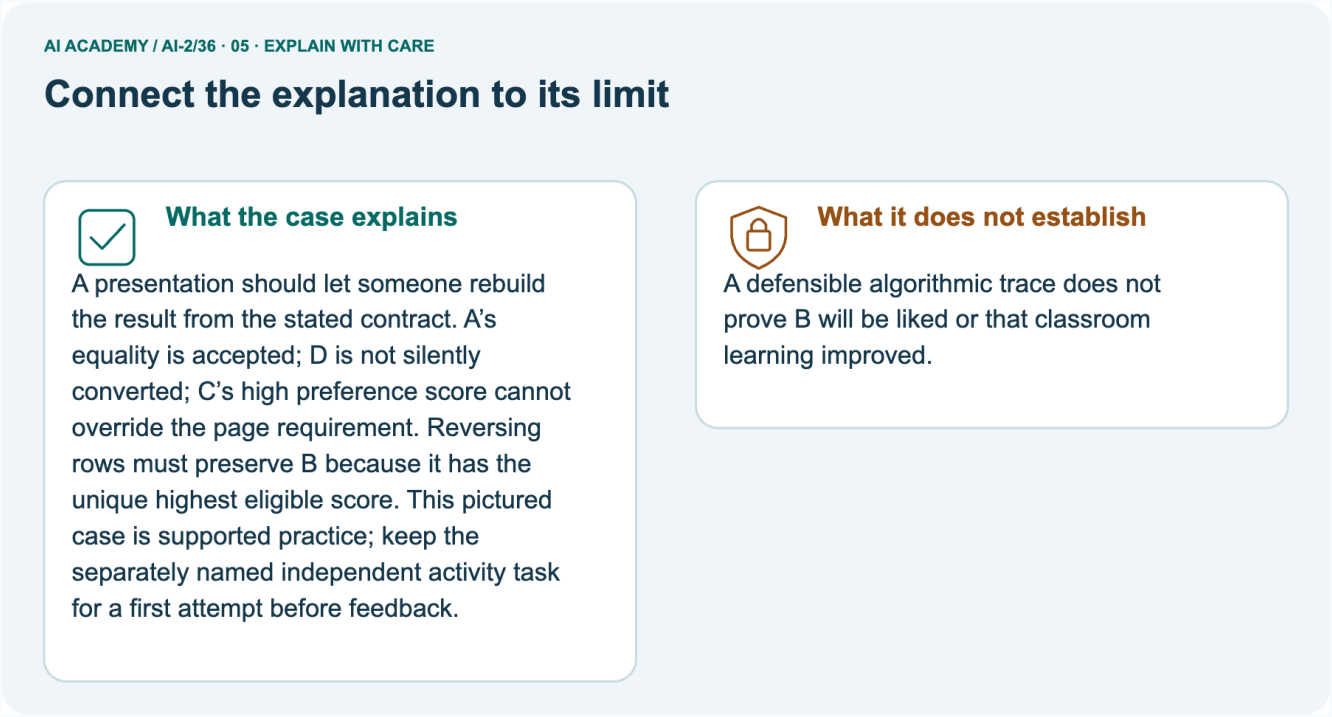
Original case: Explain two eligible items, one page-limit exclusion and one type rejection.
Changed condition: An authorised repair converts D's interest to integer 4.

Expected result: D becomes eligible and scores 9, so it wins. Reason: The **evidence**-backed **input** repair changes validation and then ranking; record this as a new case/version.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



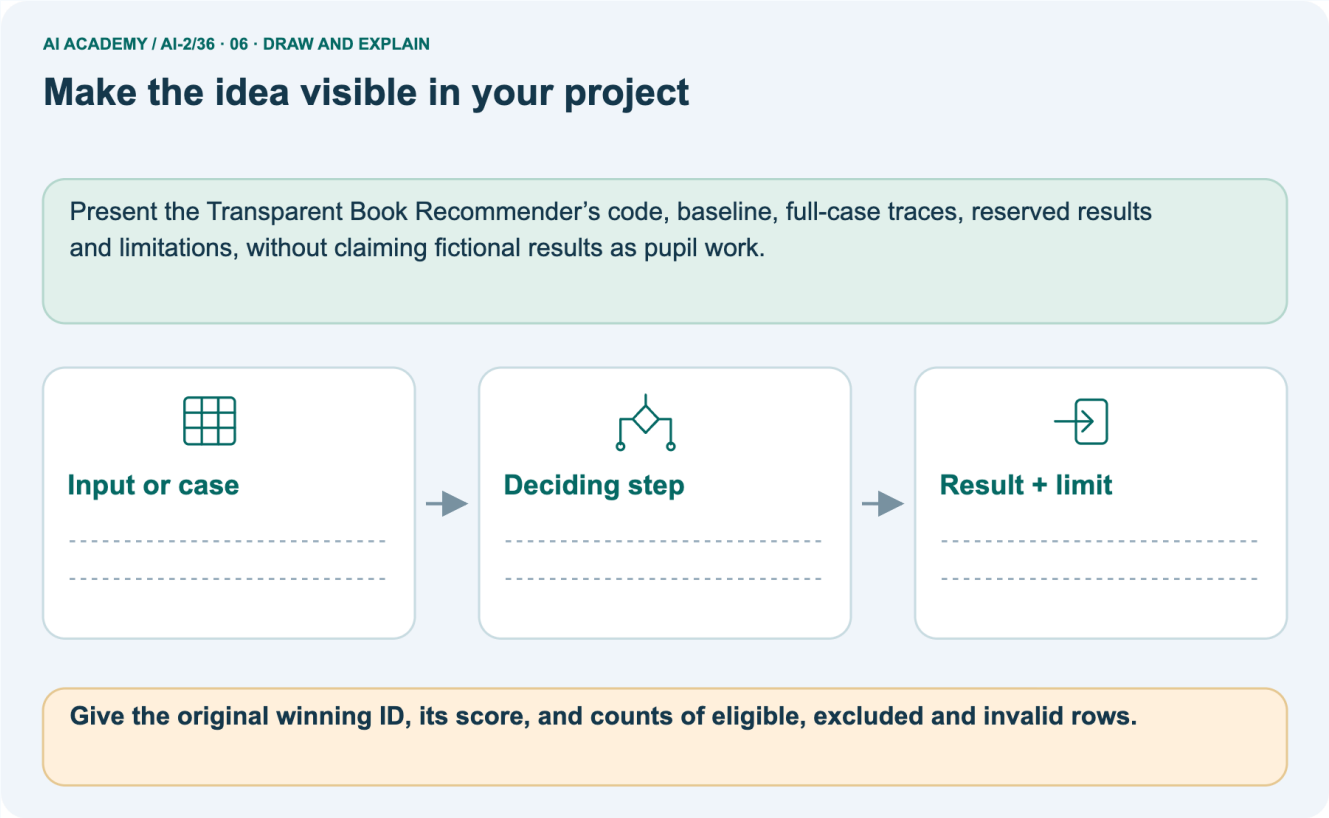
Follow the labelled picture with the matching explanation. Open the vector diagram

A presentation should let someone rebuild the result from the stated contract. A's equality is accepted; D is not silently converted; C's high preference score cannot override the page requirement. Reversing rows must preserve B because it has the unique highest eligible score. This pictured case is supported practice; keep the separately named independent activity task for a first attempt before feedback. Limit: A defensible algorithmic trace does not prove B will be liked or that classroom learning improved.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Present the Transparent Book Recommender's code, baseline, full-case traces, reserved results and **limitation**s, without claiming fictional results as pupil work. Draw your **input**, deciding step, result and limit.

Supported question: Give the original winning ID, its score, and counts of eligible, excluded and invalid rows.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Capstone Expo: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.